

First presentation

Marcel Siwiela

Objective of the project

Reading data from several sensors:

- Temperature and humidity – DHT11
- Light intensity – BH1750
- Pressure – BMP280

Data transfer of measured values in the following order:

NUCLEO-G071RB -> ESP8266 ---(WiFi)---> ESP8266 ->
-> NUCLEO-L476RG -> DISPLAY

Specifications

1. Create simple program which displays something on display on stm32-L4
2. Implement sensors on stm32-G017 in order:
 - a) DHT11
 - b) + BMP280
 - c) + BH1750
3. Implement buzzer with conditions
4. Implement WiFi module
5. Make visualisation and send info through wifi

Design must include:

1. Parallel processing
2. Schematic in KiCad
3. No user inputs

A description of hardware

- NUCLEO-G071RB
- ESP8266 x2 (**WiFi**)
- NUCLEO-L476RG
- Waveshare 4inch TFT Touch Shield
 - Temperature and humidity – DHT11
 - Light intensity x2 – BH1750
 - Pressure – BMP280
 - Wires, breadboard

Planned final result of the project.

Displaying sensor values on a screen using data transmission via
Wi-Fi modules